From Teacher Needs to Agentic AI: Designing and Validating a Personalized Career Counseling System

Bogyeom Park
Seoul National University of Science and Technology
bogyeom@seoultech.ac.kr

Mina Yoo
Chuncheon National University of Education
ymn@cnue.ac.kr

Dongkuk Lee
Kyungpook National University
dklee@knu.ac.kr

Choi Mi-ae
KERIS Digital Education Planning Section
cma1020@keris.or.kr

Seona Park
KERIS Digital Education Planning Section
sapark@keris.or.kr

So Young Jo
KERIS Digital Education Planning Section
soyoung@keris.or.kr

Kyoungwon Seo
Seoul National University of Science and Technology
kwseo@seoultech.ac.kr

Figure 3. High-fidelity 'Comprehensive Opinion' view of the agentic AI–based career and college counseling support system

Abstract

This study proposes and evaluates an agentic artificial intelligence (AI)–based system designed to support teachers' career and college counseling within the context of Korea's high school credit system. The system operationalizes a Perception–Reasoning–Action–Feedback workflow to automate the analysis of school records, integrate fragmented career-related information, and provide traceable justifications for AI-generated advice. Adopting a developmental research design, we conducted focus group interviews with in-service teachers to derive functional requirements, followed by the design and implementation of low- and high-fidelity prototypes. The system's effectiveness was then evaluated through two rounds of expert validation. Results indicate that the final five core functions—"Comprehensive Opinion," "Student Addition & File Upload," "Career & Major Information," and "Elective Course Information"—received high ratings for usefulness and content validity, while "Creative Experiential Activity Information" revealed a need for enhanced local contextualization. Overall, the

findings suggest that agentic AI can serve as a robust augmentation tool, significantly reducing administrative workload and supporting data-informed decision-making in complex counseling environments. This study establishes both a technical and pedagogical foundation for the deployment of agentic AI in real school settings.

Keyword

Agentic AI, High school credit system, Career and college counseling, Multi-agent system, Perception–Reasoning–Action–Feedback workflow

1. Introduction

The full-scale implementation of the high school credit system in 2025 marks a critical turning point in Korean high school education, shifting from a standardized national curriculum to personalized education aligned with each student's aptitude and career aspirations. While this reform empowers students to develop self-directed career planning skills, it also demands that teachers provide highly specialized counseling based on in-depth analyses of school records, elective course structures, and constantly changing college admission information. However, in practice, heavy administrative workloads and the increasing complexity of the admission system make it difficult for teachers to offer truly personalized counseling, resulting in a substantial counseling-related burden [1].

To address these issues, various career and college counseling tools have been developed, including chatbots grounded in motivational frameworks such as self-determination theory [2, 3]. Yet, existing systems remain largely limited to fragmented information retrieval or probabilistic predictions based on quantitative data. Even recently emerging generative AI–based services tend to respond passively in a question–answer format and often fail to reflect each student's individual context, leading to hallucinations. Consequently, they fall short as professional decision-support tools for complex educational counseling. Effectively supporting such decision-making requires systems that transcend simple text generation and are capable of goal recognition, multi-step planning, and autonomous use of external tools [4, 5].

Against this backdrop, the present study aims to design and validate an agentic AI–based career and college counseling support system that assists teachers' professional judgment through an autonomous cyclical workflow of Perception–Reasoning–Action–Feedback. This developmental research (1) analyzes teachers' practical difficulties and requirements for agentic AI within the high school credit system, (2) designs and develops an agentic AI reasoning workflow and corresponding prototypes based on these requirements, and (3) evaluates the educational effectiveness and practical feasibility of the final system through expert validation. The specific research questions (RQs) are as follows:

- **RQ1:** In supporting career and college counseling within the high school credit system, what practical challenges do teachers experience, and what core functional requirements do they have for an agentic AI–based system?
- **RQ2:** How are the reasoning workflow and prototypes (low-fidelity and high-fidelity) of the agentic AI system designed based on the derived requirements, and in what ways are they refined through expert feedback?
- **RQ3:** How do career and college counseling experts evaluate the educational validity and practical applicability of the final agentic AI–based counseling support system?

2. Related work

2.1 From Generative AI to Agentic AI

Recent advances in artificial intelligence (AI) are shifting the focus from generative AI, which produces content in response to user prompts, to agentic AI, which autonomously interprets goals, constructs plans, and acts within complex environments. Agentic AI is typically defined as a system that executes a sequence of actions with minimal human intervention, flexibly leveraging external tools to achieve specific goals.

The fundamental distinction between conventional generative AI and agentic AI lies in their structural workflows (Figure 1). Traditional generative AI typically follows a linear, one-shot Input–Perception–Action–Output process (Figure 1-A), where the model "perceives" a user prompt and generates a response without explicit intermediate planning. Agentic AI, however, operates through a cyclical workflow that explicitly inserts a Reasoning step between Perception and Action, utilizing a Feedback loop to iteratively refine the output (Figure 1-B) [6].

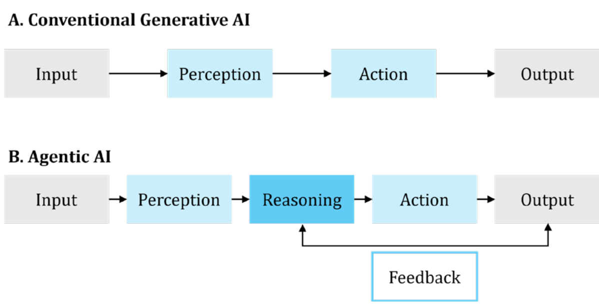

Figure 1. Conceptual comparison between conventional generative AI and agentic AI workflows

Within this cycle, each component serves a distinct purpose: Perception interprets the broader situation and context; Reasoning decomposes the overall goal into sub-tasks and formulates solution plans; Action calls external tools (e.g., APIs, information retrieval, or data manipulation) to produce concrete outputs; and Feedback detects and corrects errors or omissions to enhance completeness and reliability.

Due to this structure, agentic AI is increasingly recognized as suitable for complex decision-making that requires integrating diverse information over time, rather than merely answering isolated questions. Recent multi-agent large language model systems—which assess higher-order competencies like critical thinking through orchestrated debates—exemplify the potential of agentic architectures for complex educational tasks [7, 8]. In the domain of career and college planning, where heterogeneous information such as students' grades, school records, career aspirations, and admission policies must be holistically considered, agentic AI holds strong potential to complement and extend teachers' professional expertise.

2.2 AI-based Career and College Counseling Systems

Efforts to support students' career and college planning through data and AI are actively progressing in both research and school practice. AI-based systems in this domain have emerged as conversational tools, data-driven integrated platforms, and experiential simulation systems. While these tools offer automation and convenience, they still exhibit critical gaps when evaluated through the Perception–Reasoning–Action–Feedback workflow.

At the academic level, Shilaskar et al. (2024) proposed a conversational AI model based on natural language processing that facilitates dialogue with students to explore their interests and provide advice [8]. This work highlights the value of AI in identifying student preferences, primarily contributing to the Perception function by analyzing user utterances. However, its Reasoning capabilities—such as decomposing complex career goals and planning long-term pathways—remain limited. Furthermore, Action is largely restricted to text-based advice with minimal integration of external data such as school records, and a systematic Feedback mechanism for adapting future interaction strategies is notably absent.

At the practical level, public platforms such as *Kkumitda*, operated by the Gyeonggi Provincial Office of Education, are representative examples. Kkumitda is an integrated platform that supports counseling based on students' academic and aptitude data, leveraging large-scale admission data to provide personalized information. Such platforms perform reasonably well in Perception (processing structured academic data) and Action (presenting university and career information). However, their functions are often fragmented into separate modules, providing only limited Reasoning for contextually interpreting unstructured text data—such as school records—to organically connect career paths, majors, grades, and activities. Likewise, Feedback functions that adapt subsequent counseling based on previous outcomes are insufficient.

In sum, while existing systems partially realize Perception and Action through information retrieval and recommendation algorithms, they do not implement a full agentic workflow that independently reasons through students' complex contexts, actively explores heterogeneous data sources, and iteratively refines outcomes. Most remain at the level of "instant response to input" rather than autonomously coordinating multi-step tasks. These limitations underscore the need for an agentic AI–based counseling support system that integrates the full Perception–Reasoning–Action–Feedback cycle within the context of the high school credit system.

3. Research Methodology

This study adopted a developmental research approach to effectively support teachers' career and college counseling within the high school credit system. The overall research design was structured into three distinct phases: Phase 1 (requirements analysis), Phase 2 (design and first validation), and Phase 3 (development and second validation). Detailed descriptions of the research procedure, participants, instruments, and data analysis methods for each phase are provided in the following sections.

3.1 Research Procedure

The study proceeded in three phases. Phase 1 identified teachers' practical difficulties in career and college counseling under the high school credit system and their requirements for an agentic AI–based support system. To this end, an online focus group interview (FGI) was conducted with teachers experienced in both system operation and counseling, from which key support areas and seven major functional requirements for the agentic AI system were derived.

In Phase 2, the agentic AI reasoning workflow was designed based on these requirements, followed by the development of a low-fidelity prototype to visualize the workflow. The first round of expert validation was then conducted with specialists in career and college counseling and AI tool use to examine the appropriateness of the workflow and initial model. By integrating quantitative and qualitative feedback, we refined the logical connections among functions, finalized a design specification with five core components, and set the direction for a high-fidelity prototype.

Finally, Phase 3 involved the implementation and validation of the high-fidelity prototype. Using Azure OpenAI and the Streamlit framework, we developed a system capable of processing actual data and conducted a second round of expert validation. This phase focused on examining the educational effectiveness and practical feasibility of the final system in real school settings.

3.2 Participants

To ensure validity and practical relevance, we recruited teachers and experts with domain expertise for each research phase. All participants were informed of the study's purpose and procedures and provided voluntary consent, with all data being anonymized. For the FGI, ten high school teachers with extensive experience in the high school credit system and career counseling were selected. This group included Grade 12 homeroom teachers, full-time career guidance teachers, and heads of counseling departments, averaging over 15 years of teaching experience. Additionally, an expert validation panel was formed, consisting of four specialists with over ten years of experience who were also proficient in using large language model (LLM) tools such as ChatGPT. This panel participated in both the first (low-fidelity) and second (high-fidelity) validation rounds, ensuring that design improvements were consistently reflected. With their combined domain expertise and AI literacy, these participants supported a balanced assessment of the prototype's educational appropriateness and technical validity.

3.3 Research Instruments

To achieve the objectives of each research phase, two structured instruments were developed: an FGI interview guide and an expert validation survey. The FGI guide was designed to contrast the current career and college counseling process (AS-IS) with the envisioned process following the introduction of agentic AI (TO-BE). The interview questions were organized into three thematic categories: (1) teachers' current practices and challenges in counseling, (2) limitations of existing counseling tools, and (3) functional requirements for an agentic AI–based system.

The expert validation survey was utilized to assess the validity of the designed workflow and the functional components of the prototype. Evaluation items were organized into four dimensions—Usefulness, Ease of Use, Satisfaction, and Intention to Use—and rated on a 5-point Likert scale. Additionally, open-ended questions were included to gather detailed suggestions for improvement, enabling the simultaneous collection of quantitative evaluations and qualitative feedback.

3.4 Data Collection and Analysis

This study employed a mixed-methods approach, utilizing both qualitative and quantitative analyses depending on the research phase. For the requirements analysis phase, qualitative analysis was conducted on FGI data from ten in-service teachers. Interview recordings were transcribed and analyzed using the constant comparative method: meaning units related to counseling difficulties and agentic AI requirements were extracted, compared, and organized into hierarchical categories to derive core functional components and design directions for the system.

Subsequently, quantitative analysis was applied to expert validation survey data from both the design/first validation and development/second validation phases. Descriptive statistics (mean and standard deviation) were calculated for each item to examine overall evaluation tendencies, while the content validity index (CVI) was computed to assess agreement among experts. This process allowed for verifying the educational appropriateness of each functional component and prioritizing the refinement of components with relatively low CVI by incorporating qualitative feedback.

4. Results

4.1 Results of Phase 1: Requirement Analysis

FGI analysis revealed three main factors that hinder personalized counseling within the high school credit system: excessive teacher workload, functional limitations of existing tools, and concerns about the reliability of AI-based systems. To address these, teachers emphasized the need for "automatic analysis" of large volumes of school records, "automatic integration" of fragmented career-related information, and "clear justification" for AI-generated responses to mitigate hallucinations.

Reflecting these requirements, the career and college counseling support process was structured into two core workflows and seven functional components. The first workflow, Internal Data Analysis, corresponds to the Perception–Reasoning stages of agentic AI. It aims to reduce the time needed to understand individual students while supporting data-driven analysis through:

- **Main Dashboard**: Visualizes key student data, including grades, characteristics, and career status, in a centralized overview.

- **Student Addition & File Upload**: Enables the upload of school records (PDF/CSV), which the AI automatically recognizes and structures for analysis.

The second workflow, External Data Retrieval, corresponds to the Action–Feedback stages and focuses on integrated information provision and reliability. Here, the AI retrieves and verifies external information matched to the student's profile and generates evidence-based insights through:

- **Career Information**: Recommends occupational groups accompanied by logical justification.
- **Department & Major Information**: Presents university departments and core curricula linked to desired career paths.
- **Elective Course Information**: Suggests a tailored course selection roadmap aligned with the high school credit system.
- **Book Information**: Recommends relevant literature and related activities to strengthen career competencies.
- **Activity Information**: Recommends practical activities, such as clubs and volunteer work, to enhance career exploration.

Together, these seven components translate teachers' practical needs into concrete technical specifications. They establish a system-level foundation that allows agentic AI to move beyond simple information retrieval and substantively support teachers' professional counseling work.

4.2 Results of Phase 2: System Design and 1st Validation

Based on the seven functional components derived in Phase 1, we specified the Perception–Reasoning–Action–Feedback workflow of the agentic AI system and developed a low-fidelity prototype visualizing this workflow (Figure 2). The first round of expert validation was then conducted with specialists experienced in career counseling and AI tool use to evaluate the prototype's appropriateness.

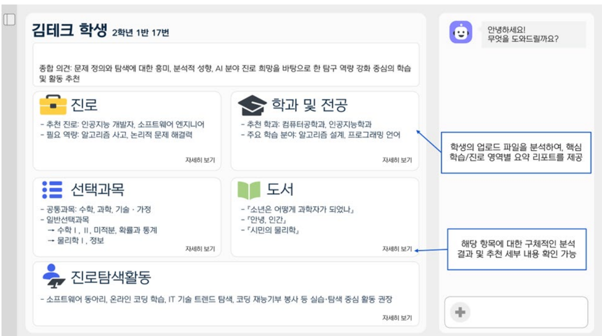

Figure 2. Low-fidelity dashboard view of the agentic AI–based career and college counseling support system

According to the quantitative results (Table 1), the overall mean score was 4.37, indicating that the design direction was perceived as highly valid. In particular, "Student Addition & File Upload" (Usefulness 5.00, CVI 1.00) and "Career Information" (Usefulness 4.75, CVI 1.00) were rated as especially effective in reducing teachers' workload.

Qualitative feedback, however, highlighted several areas for refinement. Experts noted an insufficient logical linkage among information elements and the necessity of reflecting recent policy changes. Specifically, they emphasized that "career" and "major" information should be provided in an integrated manner, while "book" information would be more appropriate as part of activity recommendations rather than an independent function, given its indirect relevance to admissions. Furthermore, they suggested upgrading the "Main Dashboard" into a Comprehensive Opinion format to analytically present each student's strengths and areas for improvement. Reflecting on these findings, the initial seven components were restructured into the following five core functional components:

- **Comprehensive Opinion**: Provides a structured summary of each student's strengths, areas for improvement, and key competencies to support holistic development.
- **Student Addition & File Upload**: Automatically recognizes and preprocesses NEIS school records (PDF/CSV), significantly reducing administrative burden.
- **Career & Major Information**: Integrates guidance by linking desired careers, related majors, core required courses, and career relevance.
- **Elective Course Information**: Suggests hierarchical course selection strategies by aligning students' career roadmaps with school offerings.
- **Creative Experiential Activity Information**: Links recommended books with autonomous, club, volunteer, and career-related activities to support competency development.

This restructuring was a critical step in evolving the system from a simple information provider into an integrated counseling tool that substantively supports teachers' professional judgment.

4.3 Results of Phase 3: System Development and 2nd Validation

Building on the five restructured functional components from Phase 2, we developed a high-fidelity prototype using Azure OpenAI (GPT-4o) and Streamlit (Figure 3). To implement the Perception–Reasoning–Action–Feedback workflow at a level applicable to real counseling scenarios, the system employs multiple specialized agents (e.g., holistic student analysis and career exploration) that collaborate autonomously.

Table 2 presents the second expert validation results for the two primary workflows: Internal Data Analysis (Perception & Reasoning) and External Data Retrieval (Action & Feedback). The overall mean score was 4.0, indicating generally positive evaluations of educational effectiveness and practical feasibility. Specifically, "Student Addition & File Upload" (Usefulness 5.0, CVI 1.00) was again confirmed as a high-value tool for reducing administrative workload. Similarly, the integrated "Career & Major Information" (Usefulness 4.75, CVI 1.00) and "Elective Course Information" (Usefulness 5.0, CVI 1.00) demonstrated very high validity. In contrast, "Creative Experiential Activity Information" received a relatively low Satisfaction score of 2.75. Qualitative feedback from experts indicated that this function did not sufficiently reflect school-specific curricula or distinctive local programs, suggesting a critical need for further localization.

Taken together, the two rounds of expert validation indicate that initial design limitations were largely resolved, establishing a solid technical and educational foundation for agentic AI as an augmentation tool for teachers' professional expertise. These results provide empirical support for both the refinement process of the high-fidelity prototype (RQ2) and the system's educational validity and practical applicability (RQ3).

5. Discussion and Conclusion

This study designed an agentic AI–based system to support teachers' career and college counseling under the high school credit system and examined its effectiveness through two rounds of expert validation. Regarding RQ1, the needs analysis identified three core requirements: automatic analysis of school records, integration of fragmented career-related information, and clear justification of AI-generated responses. These led to the derivation of two key workflows—internal data analysis and external data retrieval—and seven initial functional components. For RQ2, the first expert validation underscored the need for stronger logical linkages and the reflection of recent policy changes. This resulted in a restructuring into five core components: Comprehensive Opinion, Career & Major Information, Elective Course Information, Creative Experiential Activity Information, and Student Addition & File Upload, thereby concretizing the Perception–Reasoning–Action–Feedback workflow in an educationally appropriate form. For RQ3, the second expert validation of the high-fidelity prototype yielded generally positive evaluations regarding educational validity and practical feasibility.

Overall, the core functions of the agentic AI system proved effective in reducing teachers' workload and supporting professional decision-making. Specifically, Student Addition & File Upload and Career & Major Information consistently received high usefulness ratings. This indicates that agentic workflows—which go beyond simple text generation by autonomously leveraging external tools and real-time search—are particularly valuable for processing volatile admission data and filtering large volumes of information. The restructuring of functions further enhanced the system's educational appropriateness. These results suggest that agentic AI acts not as a replacement for teachers, but as an augmentation tool that extends their professional expertise, aligning with findings that K–12 teachers envision AI in complementary roles that preserve human pedagogical judgment [10].

While usefulness was rated highly, relatively lower Satisfaction scores indicate perceived uncertainty in some functions and concerns about responsiveness. Nevertheless, the high Intention to Use suggests that teachers view the system as a promising long-term collaborative partner. To maximize practical impact, further research is required. First, increasing satisfaction with qualitative contexts requires hyper-personalized data; future work should refine the autonomous "Action" loop by integrating school-level curricula with real-time API-based retrieval and retrieval-augmented generation (RAG) to increase contextual fit and reliability. Second, pilot studies in real school settings are needed to examine educational outcomes, such as changes in teachers' self-efficacy. Finally, incorporating predictive modeling based on longitudinal student growth data could further amplify the potential of agentic AI–based counseling support. In conclusion, this study establishes both a technical and educational foundation for an agentic AI system that substantively supports teachers within the high school credit system, providing a robust basis for future deployment and refinement.

Acknowledgement

This study was supported by the Korea Education and Research Information Service (KERIS) through the 2025 research project "A Study on Applying Agentic AI for Personalized Career and College Counseling Support."

Reference

1. Lee, D., Kim, D., Loeser, M., & Seo, K. (2025, January). How Large Language Models are Transforming Teachers' Assessment of Student Competency: A Case Study on LLM-Based Report Writing. In 2025 International Conference on Electronics, Information, and Communication (ICEIC) (pp. 1-4). IEEE.

2. Han, H., Park, B., & Seo, K. (2025, April). A Self-Determination Theory-based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (pp. 1-9).
3. Li, Y., Zhou, X., & Chiu, T. K. (2025). Systematics review on artificial intelligence chatbots and ChatGPT for language learning and research from self-determination theory (SDT): what are the roles of teachers?. Interactive Learning Environments, 33(3), 1850-1864.
4. Google Cloud. (2024). What is agentic AI? Definition and differentiators. Google Cloud.
5. IBM. (2024). What are agentic workflows? IBM Think.
6. Schneider, J. (2025). Generative to agentic ai: Survey, conceptualization, and challenges. arXiv preprint arXiv:2504.18875.
7. Park, B., & Seo, K. (2025, April). Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (pp. 1-13).
8. Jin, S. H., Im, K., Yoo, M., Roll, I., & Seo, K. (2023). Supporting students' self-regulated learning in online learning using artificial intelligence applications. International Journal of Educational Technology in Higher Education, 20(1), 37.
9. Shilaskar, S., Bhatlawande, S., Sawle, P., Gupta, S., & Buche, R. (2024, April). Conversational AI for Career Counseling. In 2024 MIT Art, Design and Technology School of Computing International Conference (MITADTSoCiCon) (pp. 1-5). IEEE.
10. Seo, K., Yoo, M., Dodson, S., & Jin, S. H. (2025). Augmented teachers: K–12 teachers' needs for artificial intelligence's complementary role in personalized learning. Journal of Research on Technology in Education, 57(4), 876-893.

Table 1. Results of the 1st Expert Validation (Low-fidelity Prototype)

Workflow	Functional Elements	Usefulness M ± SD (CVI)	Ease of Use M ± SD (CVI)	Satisfaction M ± SD (CVI)	Intention to Use M ± SD (CVI)
Internal Data Analysis	Main Dashboard	4.50 ± 0.58 (1.00)	4.25 ± 0.96 (0.75)	3.75 ± 0.96 (0.50)	4.75 ± 0.50 (1.00)
	Student Addition & File Upload	5.00 ± 0.00 (1.00)	4.50 ± 0.58 (1.00)	4.50 ± 0.58 (1.00)	5.00 ± 0.00 (1.00)
External Data Retrieval	Career Information	4.75 ± 0.50 (1.00)	4.75 ± 0.50 (1.00)	4.25 ± 0.50 (1.00)	4.50 ± 0.58 (1.00)
	Department & Major Information	4.75 ± 0.50 (1.00)	4.50 ± 1.00 (0.75)	4.25 ± 0.96 (0.75)	4.25 ± 0.96 (0.75)
	Elective Course Information	4.50 ± 0.58 (1.00)	4.50 ± 0.58 (1.00)	3.75 ± 0.96 (0.50)	4.00 ± 0.82 (0.75)
	Book Information	4.00 ± 1.41 (0.75)	4.75 ± 0.50 (1.00)	4.00 ± 1.41 (0.75)	4.00 ± 1.41 (0.75)
	Activity Information	4.25 ± 0.96 (0.75)	4.25 ± 0.96 (0.75)	4.00 ± 0.82 (0.75)	4.00 ± 0.82 (0.75)

Note. M=Mean, SD=Standard Deviation, CVI=Content Validity Index.

Table 2. Results of the 2nd Expert Validation (High-fidelity Prototype)

Workflow	Functional Elements	Usefulness M ± SD (CVI)	Ease of Use M ± SD (CVI)	Satisfaction M ± SD (CVI)	Intention to Use M ± SD (CVI)
Internal Data Analysis	Comprehensive Opinion	4.25 ± 0.96 (0.75)	4.25 ± 0.96 (0.75)	3.25 ± 0.96 (0.50)	3.75 ± 1.50 (0.50)
	Student Addition & File Upload	4.75 ± 0.50 (1.00)	4.25 ± 1.50 (0.75)	4.25 ± 1.50 (0.75)	4.25 ± 1.50 (0.75)
External Data Retrieval	Career & Major Information	4.75 ± 0.50 (1.00)	4.25 ± 1.50 (0.75)	3.50 ± 1.00 (0.75)	4.00 ± 1.41 (0.75)
	Elective Course Information	5.00 ± 0.00 (1.00)	4.25 ± 0.50 (1.00)	3.50 ± 0.58 (0.50)	4.00 ± 1.15 (0.50)
	Creative Experiential Activity Information	3.50 ± 1.29 (0.50)	4.50 ± 0.58 (1.00)	2.75 ± 0.96 (0.25)	3.25 ± 1.26 (0.25)

Note. M=Mean, SD=Standard Deviation, CVI=Content Validity Index.